University by one of its originators, Neil Turok. The second time, it was in a science fiction story about aliens. It's not often that somewhat esoteric theoretical constructions developed to solve complex problems in early universe physics show up in fiction, so it was something of a novelty at the time. The story, "Mixed Signals" by Lori Ann White and Ken Wharton, tells of a series of strange events that ultimately appear to be linked to gravitational waves. Specifically: weirdly powerful gravitational waves too regular to be due to the usual suspects—black holes or neutron stars colliding. Eventually the protagonists figure out the waves are a signal from intelligent beings, sent through the higher-dimensional bulk from *another brane*. The authors even name-check the ekpyrotic model, explaining that in this theory our universe is just one of several 3-dimensional branes in a higher-dimensional space, in which gravity but nothing else can travel. And if gravity can traverse the bulk, gravitational waves could be an excellent inter-brane communication mechanism.

While the existence of other civilizations on nearby brane-universes was never technically ruled out as a possibility, the main purpose of the hypothesis was to explain the origin and destruction of *this* universe. Not long after the lecture and the sci-fi story, I ended up doing my PhD thesis on early universe physics with Paul Steinhardt, who worked with Neil Turok to come up with the ekpyrotic model. While I focused more on other theories of our cosmic origins, I encountered the ekpyrotic scenario on a regular basis at group meetings and discussions. (Somehow, aliens never really came up.)

Since those days, the ekpyrotic scenario has been revised and generalized, and the latest version doesn't include extra dimensions at all. But as often happens in science, a novel idea that may not work in the end can still spur on a different way of thinking about the problem—one that can lead us in a totally new (and hopefully better) direction. So let's start with the original idea. It does, after all, present us with the possibility of an intriguingly dramatic cosmic end.

The term "ekpyrotic" comes from the Greek for "conflagration," a reference to the fiery origin and ultimate death of the universe in this scenario. In the standard, non-ekpyrotic story, the beginning of the universe included a period of cosmic inflation,[fn2] which we discussed in Chapter 2. Inflation causes a dramatic stretching of the cosmos during the first tiny fraction of a second, after which the decay of whatever was responsible for